

Equivalence relation and Partition

Equivalence relation

- An **equivalence relation** is a binary relation that is reflexive, symmetric and transitive.

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
<i>a</i>	✓	✓				
<i>b</i>	✓	✓				
<i>c</i>			✓			
<i>d</i>				✓	✓	✓
<i>e</i>				✓	✓	✓
<i>f</i>				✓	✓	✓

Example of equivalence relation

Example

Let A be a set of students and R be a binary relation of A such that (a, b) is in R if and only if a lives in the same dormitory as b . Since everybody lives in the same dormitory as himself or herself, R is a reflexive relation. Note that if a lives in the same dormitory as b , then b lives in the same dormitory as a . Thus, R is a symmetric relation. Note that if a lives in the same dormitory as b and b lives in the same dormitory as c , then a lives in the same dormitory as c . Thus, R is a transitive relation. Consequently, R is an equivalence relation.

Example

Ex: Let $A = \mathbb{Z}$ set of integers and let R defined by aRb if and if $a \leq b$ is R an equivalence relation,

- A **partition of a set** is a grouping of its elements into non-empty subsets, in such a way that every element is included in exactly one subset.
- Union of all subsets is equal to original set and intersection of any two subsets is Null set.

Example

- The set $\{1, 2, 3\}$ has these five partitions (one partition per item): $\{ \{1\}, \{2\}, \{3\} \}$
 - $\{ \{1, 2\}, \{3\} \}$
 - $\{ \{1, 3\}, \{2\} \}$
 - $\{ \{1\}, \{2, 3\} \}$
 - $\{ \{1, 2, 3\} \}$

Partition Example

Example

Let $A = \{a, b, c, d, e, f, g\}$

then $\{\{a\}, \{b, c, d\}, \{e, f\}, \{g\}\}$ is a partition
of A



subsets or blocks

→ also use notation $\{\bar{a}, \overline{bcd}, \overline{ef}, \bar{g}\}$
for a partition where we place an
overbar above the elements that are
in the same block.

Connection between equivalence relation and partition

→ from an equivalence relation on A we can define a partition of A so that every two elements in a block are related and any two elements in different blocks are not. This partition is said to be partition induced by equivalence relation and blocks in the partition are called equivalence classes.

Connection between equivalence relation and partition

→ conversely, from a partition of set A we can define an equivalence relation on A so that every two elements in the same block of the partition are related and any two elements in different blocks are not related.

Equivalence Class

~ An equivalence class of x is denoted by $[x]$

$$[x] = \{y \mid y \in A \text{ and } (x, y) \in R\}$$

~ $A = \{1, 2, 3, 4, 5\}$

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), \\ (1, 2), (2, 1), (4, 5), (5, 4)\}$$

~ Is R equivalence Relation?

- Eq. class

$$[1] = \{1, 2\}$$

①

$$[4] = \{4, 5\}$$

③

$$[2] = \{2, 1\}$$

$$[5] = \{5, 4\}$$

$$[3] = \{3\}$$

②

$$P_1 = \{1, 2\}$$

$$P_2 = \{3\}$$

$$P_3 = \{4, 5\}$$

- If set A contains n elements, then How many minimum and maximum No. of equivalence classes are possible?

Which of the following collections of subsets are partitions of $\{0, 1, 2, 3, 4, 5\}$?

1 $\{0, 1, 2\}, \{4, 3\}, \{5, 4\}$

2 $\{\}, \{0, 2, 1\}, \{4, 3, 5\}$

3 $\{5, 4, 0, 3\}, \{2\}, \{1\}$

4 $\{5\}, \{4, 3\}, \{0, 2\}$

5 $\{2\}, \{1\}, \{5\}, \{3\}, \{0\}, \{4\}$

List all the partitions of the following sets:

1 $A = \{1, 2\}$

2 $B = \{1, 2, 3\}$

Solution.

1 The set $A = \{1, 2\}$ has 2 partitions:

$$\{1\}, \{2\}$$

$$\{1, 2\}$$

2 The set $B = \{1, 2, 3\}$ has 5 partitions:

$$\{1\}, \{2\}, \{3\}$$

$$\{1, 2\}, \{3\}$$

$$\{1, 3\}, \{2\}$$

$$\{1\}, \{2, 3\}$$

$$\{1, 2, 3\}$$

Example 3.

List the ordered pairs in the equivalence relation R induced by the partition $P = \{\{a, b, c\}, \{d\}, \{e\}\}$ of the set $\{a, b, c, d, e\}$.

Solution.

The partition P includes 3 subsets which correspond to 3 equivalence classes of the relation R . We can denote these classes by E_1 , E_2 , and E_3 . They contain the following pairs:

$$E_1 = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$$

$$E_2 = \{d, d\}$$

$$E_3 = \{e, e\}$$

So, the relation R in roster form is given by

$$R = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c), (d, d), (e, e)\}.$$

A relation R on the set $A = \{a, b, c, d, e\}$ is defined as follows: $R = \{(a, a), (a, b), (b, a), (b, b), (c, c), (c, d), (c, e), (d, c), (d, d), (d, e), (e, c), (e, d), (e, e)\}$. Determine the equivalence classes of R .

- **Show that the relation “Congruence modulo m over the set of positive integers is an equivalence relation”**

Congruence modulo

Definition: Two integers a and b are congruent modulo m iff they have the same remainder when divided by m .

denoted by: $a \equiv b \pmod{m}$

read as: a is congruent to b modulo m .

Note:

>> $a \equiv b \pmod{m}$ means $a \bmod m = b \bmod m$.

>> $a \equiv b \pmod{m}$ iff m divides $a - b$.

Congruence modulo

Ex

$$16 \equiv 13 \pmod{3}$$

$$\sim 16 \bmod 3 = 1$$

$$13 \bmod 3 = 1$$

also,

$$3 \text{ divides } 16 - 13 = 3$$

~ other examples

$$29 \equiv 8 \pmod{7}$$

$$60 \equiv 0 \pmod{15}$$